

Software Engineering Lab - Lab 2

Agile Backlog Creation & Sprint Simulation in Jira

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Problem Statement #48 - Incident Escalation & On-Call Rotation Engine

1. Epics and User Stories

Epic 1: On-Call Rotation & Schedule Management

Epic 2: Alert Ingestion & Verification Pipeline

Epic 3: Notification & Escalation Engine

Epic 4: Multi-Channel Triage & Incident Lifecycle

Epic 5: Incident Post-Mortem & Reporting

Epic	Issue	User Story	Points	Priority
Epic 1	SD-6	Recurring Shift Rotations	5	Medium
Epic 1	SD-7	Shift Overrides & Holiday Swaps	3	Medium
Epic 2	SD-8	Webhook Alert Ingestion	5	High
Epic 2	SD-9	Active Responder Resolution	3	High
Epic 3	SD-10	Escalation Ladder Configuration	5	High
Epic 3	SD-11	5-Minute Escalation Trigger	8	High
Epic 4	SD-12	Web Dashboard Triage	5	High
Epic 4	SD-13	Remote SMS and Voice Triage	5	High
Epic 5	SD-14	Post-Mortem Template Generation	5	Medium
Epic 5	SD-15	Escalation Exhaustion Alerting	3	Medium

Estimation: Story points use the Fibonacci scale and priorities were set as High/Medium based on dependency and user impact.

Sprint 1 plan: SD-6, SD-7, SD-8, SD-9 and SD-12 were selected, for a total of 21 story points.

2. Jira Screenshots

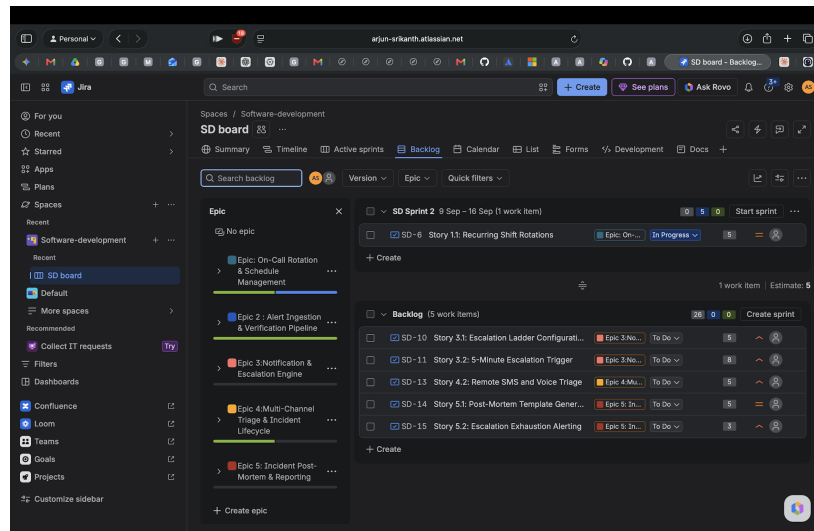


Figure 1. Backlog view: five Epics, story-point values, Sprint 2 carry-over and remaining backlog items.

Observation: The screenshot shows all five Epics. Since completed Sprint 1 items no longer appear in the backlog, the full list of ten stories and their points is kept on page 1.

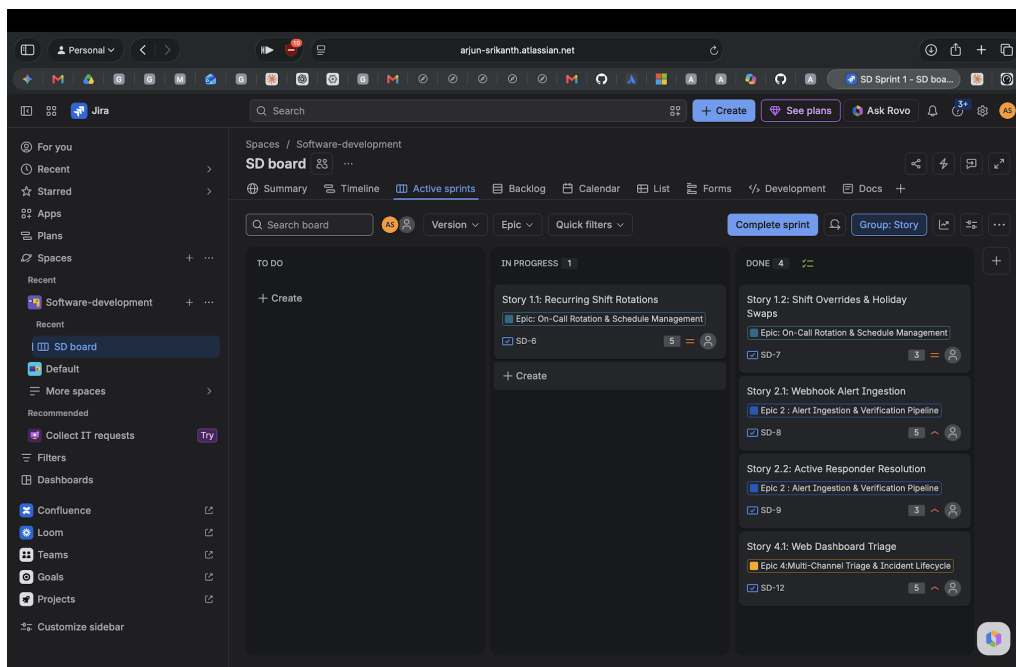


Figure 2. Active Sprint view: four stories are Done and SD-6 remains In Progress.

Observation: Done: SD-7 (3), SD-8 (5), SD-9 (3), SD-12 (5) = 16 points. In Progress: SD-6 = 5 points.

3. Burndown Chart and Sprint Result

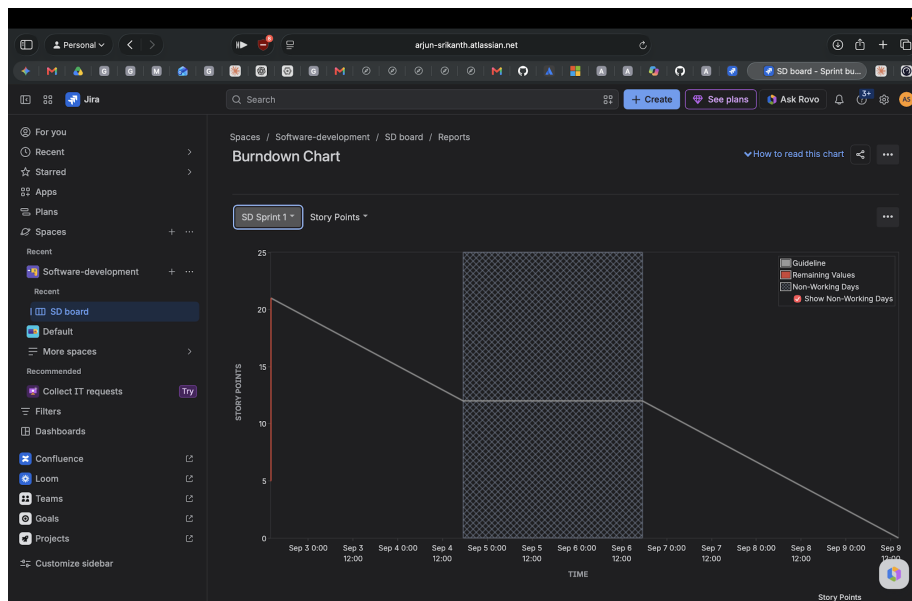


Figure 3. Burndown chart for SD Sprint 1 using Story Points.

Item	Result
Sprint 1 planned	21 story points
Completed	16 story points (SD-7, SD-8, SD-9, SD-12)
Remaining / carried forward	5 story points (SD-6)
Sprint 2	SD-6 is shown under SD Sprint 2 in the backlog screenshot

Chart reading: Remaining work drops from about 21 points to 5 points after the simulated status updates. The sharp drop is because several stories were moved to Done close together.

4. Reflection Questions

1. Did your estimations reflect the actual effort?

Mostly. The 3-point stories such as shift overrides and responder lookup were completed, but SD-6 took longer than expected. I would either give it more points or split it into smaller stories.

2. Was your backlog well-prioritized?

Yes. Alert ingestion, on-call scheduling and web triage were given higher priority because later escalation and reporting features depend on them.

3. How did your simulated sprint align with your plan?

We planned 21 points. Four stories worth 16 points were completed and SD-6, worth 5 points, was moved to Sprint 2. So we completed 16 out of 21 planned points.

4. What insights did the burndown chart give about your team's capacity?

Around 16 story points looks more realistic for a similar sprint. The sharp drop also shows that several items were updated together during the simulation.